

Hess's Law and Enthalpies of Formation

Answer key with worked solutions

A. Understanding

1. Hess's law.

The enthalpy change of a reaction is **the same whatever route** is taken from reactants to products. It is true because enthalpy is a **state function** (it depends only on the initial and final states), which follows from the conservation of energy.

2. $\Delta H_f^\circ = 0$.

O₂(g), N₂(g), Fe(s), C(graphite): elements in their standard states. O₃(g) is not the standard state of oxygen; H₂O(l) and CO₂(g) are compounds.

B. Combining equations

3. Nitrogen dioxide.

(a) The equations add directly; 2NO cancels: $\Delta H = +182.6 \text{ kJ} + (-116.2 \text{ kJ}) = \mathbf{+66.4 \text{ kJ}}$. (b) This equation forms 2 mol of NO₂ from its elements, so $\Delta H_f^\circ(\text{NO}_2) = +66.4 \text{ kJ} \div 2 \text{ mol} = \mathbf{+33.2 \text{ kJ/mol}}$.

C. Using enthalpies of formation

4. Three reactions.

(a) $\Delta H^\circ = (1 \text{ mol})(-84.0 \text{ kJ/mol}) - [(1 \text{ mol})(+52.4 \text{ kJ/mol}) + 0 \text{ kJ}] = \mathbf{-136.4 \text{ kJ}}$
(b) $\Delta H^\circ = [(1 \text{ mol})(-634.9 \text{ kJ/mol}) + (1 \text{ mol})(-393.5 \text{ kJ/mol})] - (1 \text{ mol})(-1207.6 \text{ kJ/mol}) = -1028.4 \text{ kJ} + 1207.6 \text{ kJ} = \mathbf{+179.2 \text{ kJ}}$ (endothermic: limestone must be heated strongly)
(c) $\Delta H^\circ = [(1 \text{ mol})(-1675.7 \text{ kJ/mol}) + 0 \text{ kJ}] - [(1 \text{ mol})(-824.2 \text{ kJ/mol}) + 0 \text{ kJ}] = \mathbf{-851.5 \text{ kJ}}$ (very exothermic: it melts the iron)

5. Propane.

Products: $(3 \text{ mol})(-393.5 \text{ kJ/mol}) + (4 \text{ mol})(-285.8 \text{ kJ/mol}) = -1180.5 \text{ kJ} - 1143.2 \text{ kJ} = -2323.7 \text{ kJ}$

Reactants: $(1 \text{ mol})(-103.8 \text{ kJ/mol}) + (5 \text{ mol})(0 \text{ kJ/mol}) = -103.8 \text{ kJ}$

$\Delta H^\circ = -2323.7 \text{ kJ} - (-103.8 \text{ kJ}) = \mathbf{-2219.9 \text{ kJ}}$

6. Ammonia oxidation.

Products: $(4 \text{ mol})(+91.3 \text{ kJ/mol}) + (6 \text{ mol})(-241.8 \text{ kJ/mol}) = 365.2 \text{ kJ} - 1450.8 \text{ kJ} = -1085.6 \text{ kJ}$

Reactants: $(4 \text{ mol})(-45.9 \text{ kJ/mol}) + (5 \text{ mol})(0 \text{ kJ/mol}) = -183.6 \text{ kJ}$

$\Delta H^\circ = -1085.6 \text{ kJ} - (-183.6 \text{ kJ}) = \mathbf{-902.0 \text{ kJ}}$

7. Vaporization of water.

$\Delta H^\circ = (1 \text{ mol})(-241.8 \text{ kJ/mol}) - (1 \text{ mol})(-285.8 \text{ kJ/mol}) = \mathbf{+44.0 \text{ kJ}}$. This is the **enthalpy of vaporization** of water at 25 °C: the heat needed to turn 1 mol of liquid water into vapour.

D. Challenge

8. ΔH_f° of ethanol.

$\Delta H^\circ = [2\Delta H_f^\circ(\text{CO}_2) + 3\Delta H_f^\circ(\text{H}_2\text{O(l)})] - [\Delta H_f^\circ(\text{ethanol}) + 0]$, so

$\Delta H_f^\circ(\text{ethanol}) \times (1 \text{ mol}) = (2 \text{ mol})(-393.5 \text{ kJ/mol}) + (3 \text{ mol})(-285.8 \text{ kJ/mol}) - (-1366.8 \text{ kJ})$
 $= -787.0 \text{ kJ} - 857.4 \text{ kJ} + 1366.8 \text{ kJ} = -277.6 \text{ kJ}$, so $\Delta H_f^\circ(\text{ethanol}) = \mathbf{-277.6 \text{ kJ/mol}}$ (matching the table)

9. Glucose.

Products: $(6 \text{ mol})(-393.5 \text{ kJ/mol}) + (6 \text{ mol})(-285.8 \text{ kJ/mol}) = -2361.0 \text{ kJ} - 1714.8 \text{ kJ} = -4075.8 \text{ kJ}$

$\Delta H^\circ = -4075.8 \text{ kJ} - (1 \text{ mol})(-1273.3 \text{ kJ/mol}) = \mathbf{-2802.5 \text{ kJ}}$ (per mole of glucose)

Energy from 10.0 g: $\frac{10.0 \text{ g}}{180.16 \text{ g/mol}} = 0.05551 \text{ mol}$; $0.05551 \text{ mol} \times 2802.5 \text{ kJ/mol} = \mathbf{156 \text{ kJ}}$ released